MIL-A-60148A(MU)
11 July 1966
SUPERSEDING
MIL-A-60148(MU)
7 January 1965

MILITARY SPECIFICATION

AMPLIFIER: 10527249

1. SCOPE

1.1 This specification covers one type of amplifier used in Anti-oscillation Sighting System XM58 employed in a UH-1B Helicopter. This unit is the power amplifying segment of the amplifier which energizes the azimuth and elevation control motors (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communication, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10527249	Amplifier

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527249 Packaging of Amplifier: 10527249

(Copies of specifications, standards, and drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The amplifier shall be fabricated in accordance with Drawing F10527249 and drawings pertaining thereto.

3.2.1 Treatment, moisture and fungus resistant.- The amplifier shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperatures.- The amplifier shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to temperatures at -80° and at +160°F.

3.4.2 Operating temperatures.- The amplifier shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150°F.

3.4.3 Altitude.- The amplifier shall meet the performance requirements of 3.5 at an altitude of 30,000 feet (ft) above sea level.

3.4.4 Shock.- The amplifier shall withstand impact shocks up to 15 G's, each shock impulse having a time duration of 11 ± 1 milliseconds (ms). Subsequently, the amplifier shall meet the performance requirement of 3.5.

3.4.5 Vibration.- The amplifier shall meet the performance requirements of 3.5 after being subjected to vibrations up to 5 G's over the frequency range of 5 to 500 cps.

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3.5 Performance.

3.5.1 Null.- Null value measured at the loaded output shall be equal to or less than ±1.0 volt direct current (Vdc) for a zero Vdc input signal, having a source impedance of 2200 ohms.

3.5.2 DC gain.- DC gain shall be 20 ± 2 $\frac{VDC}{VDC}$ for output voltages from +19Vdc to -19Vdc.

3.5.3 Sensing.- Positive output voltage shall be generated by a positive dc voltage at the input and a negative input shall generate a negative output.

3.5.4 Linearity.- Linearity shall be within ± 10% for all measured outputs from +19Vdc to -19Vdc.

3.6 Radio interference suppression.- Interference control requirements shall be in accordance with MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall

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be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Altitude	3.4.3	4.6.2.3
302.	Shock	3.4.4	4.6.2.4
303.	Vibration	3.4.5	4.6.2.5
304.	Radio interference suppression	3.6	4.6.4

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Secdion 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE II - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.5% Defective		
101.	Null	3.5.1	4.6.3.1
102.	DC Gain	3.5.2	4.6.3.2
103.	Sensing	3.5.3	4.6.3.2
104.	Linearity	3.5.4	4.6.3.2
105.	Moisture and fungus resistance	3.2.1	4.6.1
MINOR:	NONE		

NOTE: The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three amplifiers shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.2	Applicable drawing - visual
402.	General specification	3.3	MIL-F-13926 - visual
403.	Temperature (storage)	3.4.1	4.6.2.1
404.	Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The amplifier shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 Storage temperature.- The amplifier, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procuedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the amplifier shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperature.- The amplifier shall be subjected to the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the amplifier shall meet the performance requirements specified in 3.5.

4.6.2.3 Altitude test.- The amplifier shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level).

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After sufficient time has elapsed to allow all entrapped air to escape, the amplifier shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The amplifier shall be tested in accordance with MIL-F-13926, Procedure II. The amplifier shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the amplifier mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the amplifier shall be closely inspected for any evidence of mechanical failure, loose or broken parts. At end of test, the amplifier shall meet the performance requirements of 3.5.

4.6.3 Performance.

4.6.3.1 Null test.- Use the test set-up shown in Figure 1. With power applied to the amplifier, the amplifier output voltage shall indicate a null reading on M-2 between terminals B and S within the limits specified in 3.5.1.

4.6.3.2 Gain, linearity and sensing test.- Using the test set-up in Figure 1, apply 0, +0.2, +0.4, +0.6, +0.8 +1.0 and +1.2Vdc to the input. Repeat for 0, -0.2, -0.4, -0.6, -0.8, -1.0 and -1.2Vdc. Measure and plot the generated voltage versus each input voltage. Observe that the positive voltage inputs generate a positive voltage output, and that the negative voltage inputs generate negative voltage outputs in compliance with 3.5.3. Establish the best straight line for all output voltages between +19 and -19Vdc. Verify gain for compliance with 3.5.2 and linearity for compliance with 3.5.4.

4.6.4 Radio interference test.- The equipment shall be tested in accordance with MIL-I-6181 to insure conformance with 3.6.

5. PREPARATION FOR DELIVERY

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5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527249. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Amplifier 10527249 accepts amplified signals from Amplifier 10527248, further amplifies these by a factor of 20 and generates sufficient power to drive a 40 watt control motor that drives the gimbal of the Sight Assembly 10527236.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging, and packing (see 5.1)
c. Applicable packaging data sheet number (see 5.1)
d. Applicable stock number

Custodian:

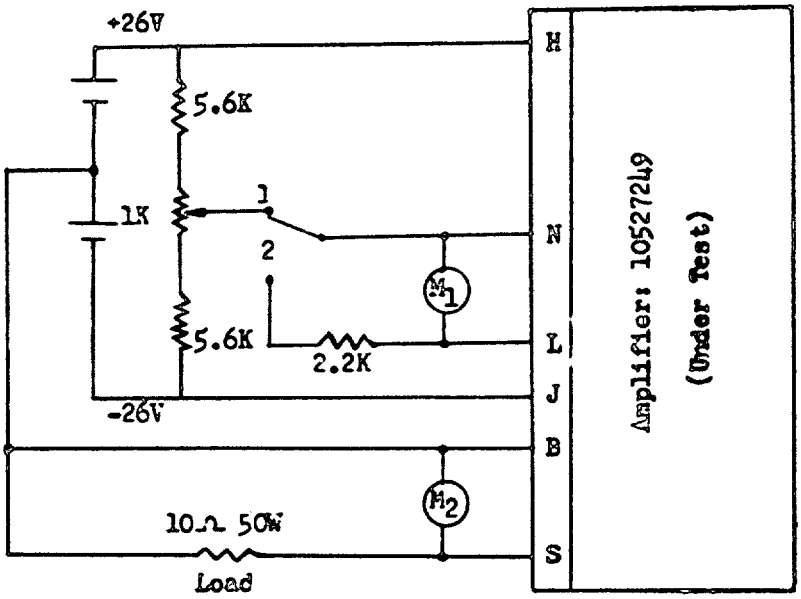
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Project No. 1270-A018Y

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NOTES:

1. Switch in position 1 for gain and linearity test.
2. Switch in position 2 for null test.
3. M_1 - Input voltmeter ±2V dc range.
4. M_2 - Output voltmeter ±30V dc range.

FIGURE 1. Test Setup for Amplifier: 10527249

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel of the Government or contractor, involved in the use of this specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that adequate products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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